Do Machine Learning Trading Strategies Survive Transaction Costs? Evidence from Cryptocurrency Markets

Group Members:

Name	Matriculation Number
Lau Ming Jie	A0272632H
Loo Wen Wen	A0259594J
Low Ruo Fei Evie	A0259605X
Wilbert Christopher Kurniawan	A0282267X
Wu Xuande	A0287897W

GitHub Repository: https://github.com/xdewu0508/QF4211Project

1. Introduction **3**
2. Literature Review **3**
3. Data Description and Processing **4**
3.1. Data Source and Sample Construction 4
3.2. Feature Construction 5
3.3. Target Definition and Data Splitting 5
4. Methodology **6**
4.1. Prediction Framework 6
4.2. Strategy Classes 6
4.3. Backtesting Framework 7
4.4. Performance Evaluation 7
5. Results and Analysis **8**
5.1. Passive Benchmark Performance 8
5.2. Active Strategy Performance After Costs 9
5.3. Strategy Direction and Robustness 9
5.4. Transaction Costs Substantially Reduce Performance 10
5.5. Statistical Usefulness of Active Strategies 11
5.6. Feature Representation and Technical Indicators 11
5.7. Equity Curve Evidence 12
5.8. Overall Interpretation 12
6. Conclusion, Limitations, and Future Work **13**
6.1. Conclusion 13
6.2. Limitations and Future Work 13
Appendix A: Core Empirical Results **15**
Appendix B. Supporting Results **22**
Appendix C. Equity Curve Analysis **26**
Appendix D. Regime-Shift Robustness Analysis **30**
Rebuttal to Reviewer Comments **36**
References **39**

1. Introduction

Recent advances in machine learning (ML) have led to growing interest in their application to financial markets, particularly in trading strategy development. In cryptocurrency markets, where price dynamics are highly volatile, ML models have been shown to extract predictive signals from historical data and technical indicators, sometimes outperforming traditional approaches in forecasting short-term price movements.

However, most existing studies evaluate performance under simplified assumptions that ignore transaction costs and execution frictions. This limits the economic relevance of reported results, especially in high-frequency settings where trading costs accumulate rapidly.

This paper asks **whether machine-learning-based cryptocurrency trading strategies still deliver economic value after transaction costs,** and answers that question by comparing them with rule-based strategies and a passive buy-and-hold benchmark. We evaluate ML and rule-based strategies under a common backtesting framework across Bitcoin, Ethereum, XRP, and Solana, and benchmark their performance against buy-and-hold. Transaction costs and execution constraints are explicitly incorporated, and performance is assessed using risk-adjusted metrics.

We find that while some ML models perform well in frictionless settings, their advantage narrows substantially once transaction costs are applied. High-turnover strategies are affected the most, while long-only strategies are more robust. Overall, ML strategies can outperform rule-based strategies in selected settings, but they do not deliver broad or consistent outperformance over buy-and-hold after costs.

2. Literature Review

The application of machine learning (ML) in financial markets has shown that flexible models can capture nonlinear relationships and improve return prediction. Gu et al. (2020)

demonstrate the strong predictive performance of neural networks in equity markets, while recurrent architectures have been widely applied to time-series forecasting.

In cryptocurrency markets, ML-based trading strategies have received particular attention due to high volatility and continuous trading. Goutte et al. (2023) show that deep learning models, especially GRU architectures, can generate profitable trading signals using technical indicators and candlestick-based features.

However, most existing studies evaluate performance under frictionless assumptions that exclude transaction costs and execution constraints. This is particularly problematic in high-frequency settings, where trading costs can substantially erode returns. As noted by Goutte et al. (2023), the omission of transaction costs limits the practical relevance of reported performance.

This paper builds directly on Goutte et al. (2023) by incorporating transaction costs into a fee-aware backtesting framework for cryptocurrency trading strategies. More importantly, it evaluates whether ML-based strategies retain economic value relative to rule-based strategies and a passive buy-and-hold benchmark after costs are applied. This allows for a more direct assessment of whether predictive performance translates into economically meaningful trading outcomes.

3. Data Description and Processing

3.1. Data Source and Sample Construction

The dataset is constructed from publicly available hourly candlestick data obtained from the Binance exchange. We focus on four liquid USDT-denominated spot trading pairs: BTC/USDT, ETH/USDT, XRP/USDT, and SOL/USDT. These assets were selected to provide a range of market characteristics, from large-cap cryptocurrencies such as Bitcoin and Ethereum to higher-volatility assets such as XRP and Solana, while maintaining a consistent trading venue and data structure across the analysis.

The sample period spans January 2021 to December 2025, yielding approximately 43,000 hourly observations per asset after feature alignment. This period captures multiple market regimes, including expansionary and contractionary phases, allowing for a more comprehensive evaluation of model performance under varying market conditions. Hourly data are used to remain consistent with prior studies while preserving an intraday trading horizon.

3.2. Feature Construction

We construct four feature sets to evaluate the impact of data representation and technical indicators on model performance. The `ohlc_raw` feature set consists of standard price and activity variables, including open, high, low, close, volume, and trade count, along with cyclical time features. The `candle_raw` set replaces the raw price variables with candlestick representations, capturing body and shadow characteristics while retaining the remaining features.

To assess the role of technical indicators, we extend both representations to form `ohlc_extended` and `candle_extended` feature sets. These include commonly used technical indicators such as moving averages, momentum indicators, and oscillators. All features are constructed using only contemporaneously available information and are appropriately lagged to ensure that no future information is used in generating trading signals.

3.3. Target Definition and Data Splitting

The prediction task is formulated as a binary classification problem, where the target variable indicates the direction of the next-hour return. This setup aligns with the objective of generating trading signals rather than predicting exact price levels.

To ensure a realistic evaluation, the data are split chronologically into training (50%), validation (25%), and test (25%) sets. This design prevents look-ahead bias and ensures that model performance is assessed on genuinely out-of-sample data. The validation set is used

for model selection and threshold tuning, while the test set is reserved for final performance evaluation.

4. Methodology

4.1. Prediction Framework

The predictive task is formulated as a binary classification problem based on the next-hour open-to-open return. At each time step, the model observes the current feature vector and generates a probability estimate of the direction of the next-hour return. A trade is executed at the next hour's open if the predicted probability exceeds a specified threshold, and the position is closed one hour later at the following open.

Two directional labels are constructed: a **positive-return label for long signals** and a **negative-return label for short signals**. This dual-labelling framework allows long-only, short-only, and long-short trading strategies to be implemented consistently across both machine learning and rule-based approaches. Importantly, model training is separated from trade execution. Models are trained to optimize classification performance, while trading decisions are determined through a thresholding procedure applied on validation data.

4.2. Strategy Classes

We evaluate three classes of strategies: a passive benchmark, rule-based technical strategies, and ML models.

The **buy-and-hold benchmark** maintains a fully invested long position throughout the test period and serves as a reference for passive performance.

For **rule-based strategies**, we consider standard technical heuristics based on moving averages (SMA) and relative strength index (RSI). These rules generate continuous signals, which are mapped into trading decisions using the same thresholding framework applied to machine learning models.

For **ML strategies**, we consider four model classes. Logistic regression provides a linear baseline, while XGBoost and LightGBM capture nonlinear feature interactions through tree-based ensembles. A gated recurrent unit (GRU) model is used to capture temporal dependencies, with a fixed lookback window of 24 hours. All models are trained using binary cross-entropy loss and output probability scores, which are subsequently converted into trading signals through thresholding. No trading objective, such as Sharpe ratio, is used during model training.

4.3. Backtesting Framework

Backtesting is conducted under two regimes: a frictionless setting (no cost) and a realistic setting (with cost). In the with-cost regime, a transaction cost of 0.035% is applied per unit change in position. Opening or closing a position incurs a cost of 0.035%, while directly reversing from long to short incurs a cost of 0.07%.

Positions are scaled to unit exposure, taking values of +1 (long), 0 (flat), or −1 (short), and are evaluated from a normalized initial capital of 1. This design isolates the effect of trading frequency and signal quality, as capital utilisation is driven by time-in-market rather than leverage.

Predicted probabilities are converted into trading signals using thresholds selected on the validation set. Threshold selection follows a precision-first rule, subject to feasibility constraints, including a minimum number of trades, minimum signal frequency, and minimum recall. For long-short strategies, if both long and short signals are triggered simultaneously, the strategy selects the side with the larger confidence margin.

4.4. Performance Evaluation

Performance is evaluated using both return-based and classification-based metrics. Following the reference paper, the primary performance measure is based on compounded signal returns. Total return is defined as

$$Total\ Return = \prod_{i=1}^{N}(1 + R_i S_i),$$

where R_i is the realized return at time i, and $S_i \in \{0, 1\}$ indicates whether a position is active. The average return is computed as

$$Average\ Return = \frac{Total\ Return}{\sum_{i=1}^{N} S_i},$$

and the Sharpe ratio is defined as

$$Sharpe\ Ratio = \frac{Average\ Return}{\sigma(R)},$$

where $\sigma(R)$ denotes the standard deviation of returns. This formulation reflects the quality of active trading signals rather than unconditional returns.

As a secondary diagnostic, we also report the annualized Sharpe ratio, computed from hourly net returns. In the implementation, the risk-free rate is set to zero, and annualization is based on 8,760 trading hours per year. Unless otherwise stated, Sharpe ratios reported in the main results refer to the test Sharpe computed from compounded signal returns, rather than annualized hourly returns.

In addition to these metrics, we report cumulative return, maximum drawdown, turnover, and number of trades to provide a comprehensive evaluation of both statistical performance and economic feasibility.

5. Results and Analysis

5.1. Passive Benchmark Performance

Table A1 reports the buy-and-hold benchmark over the test period. Passive outcomes differ sharply across assets. XRP delivers the strongest benchmark performance, with an annualized Sharpe ratio of 1.40 and cumulative return of 201.3%, while BTC also performs strongly with a Sharpe ratio of 0.82. ETH exhibits more moderate performance, and SOL records a negative cumulative return despite high volatility.

This dispersion establishes a non-trivial benchmark for active strategies. In assets such as BTC and XRP, where passive exposure already delivers strong risk-adjusted returns, active strategies must generate sufficiently large gains to justify additional turnover and transaction costs.

5.2. Active Strategy Performance After Costs

Table A2 reports the best feasible active configurations evaluated under transaction costs. The strongest results are concentrated in SOL and XRP. For example, the GRU long-short strategy using candle_extended features on SOL achieves a test Sharpe of 0.199, while the GRU long-only variant reaches 0.159. In XRP, the best performing configuration is a GRU long-only strategy with a test Sharpe of 0.136.

These results indicate that ML models can extract economically meaningful signals in selected assets. However, the magnitude of these Sharpe ratios remains modest relative to passive benchmarks. In BTC and ETH, active strategies perform substantially worse: BTC's best long-only configuration achieves a test Sharpe of 0.039, while ETH's best long-only strategy is negative at −0.011. In both cases, active strategies fail to improve on passive performance (Table A1).

Overall, the evidence suggests that while predictive signals exist, their economic value is limited once transaction costs are incorporated, particularly in assets with stronger baseline efficiency or weaker signal structure.

5.3. Strategy Direction and Robustness

Table A3 reports average performance across feasible configurations by model and strategy mode using test Sharpe. Long-only strategies consistently outperform other strategy classes. GRU long-only strategies achieve the highest average test Sharpe at 0.053, followed by Logistic Regression long-only at 0.038 and SMA long-only at 0.027. By contrast, average

short-only Sharpe is negative across all models, and long-short performance is generally weaker than long-only.

This pattern reflects both market directionality and trading frictions. Short-only strategies are consistently weak: every short-only model in Table A3 has a negative test Sharpe, even though several achieve precision slightly above 50%. This suggests that correctly predicting some downward moves is not enough to overcome the broader upward drift in major cryptocurrencies and the costs incurred from repeated short entries and exits. Long-only strategies benefit from aligning with this upward bias and generally require fewer position reversals, while short-only and long-short strategies face a larger turnover burden. As a result, even when classification precision exceeds 50%, realized Sharpe can remain negative if the average return per trade is small relative to transaction costs.

The relevant constraint is therefore not predictive accuracy alone, but the size of the tradable edge after costs. Models that generate many marginal signals may perform well statistically but poorly economically.

5.4. Transaction Costs Substantially Reduce Performance

Table A4 quantifies the impact of transaction costs. Cost drag is economically large across models and strategy modes, particularly for cumulative returns. For example, XGBoost long-short strategies experience an average cumulative return reduction of 0.976, while Logistic Regression long-short strategies lose 0.524. Even rule-based RSI strategies exhibit substantial cost drag despite relatively simple signal structures.

These magnitudes indicate that transaction costs are not a marginal adjustment but a central determinant of profitability. In many cases, cost drag exceeds the average return per trade, implying that strategies with statistically valid signals may not be economically tradable.

This effect is illustrated at the configuration level in Table B2. Several strategies that appear profitable under frictionless assumptions become flat or negative once costs are applied. This highlights the risk of overstating performance when transaction costs are ignored, particularly for strategies that rely on frequent small trades.

5.5. Statistical Usefulness of Active Strategies

The bootstrap tests in Table A5 provide a stricter assessment of whether the best active strategies generate statistically meaningful returns. Most strategies fail to reject the null of zero mean net return at conventional significance levels. The main exception is the SOL GRU long-short strategy, which has a positive mean net return and p-value of 0.0106 against zero.

However, even this strategy does not significantly outperform buy-and-hold, with a p-value of 0.4647 relative to the passive benchmark. Similarly, XRP GRU long-only has a relatively low p-value against zero return at 0.0709, but performs poorly relative to buy-and-hold. This reinforces the distinction between generating positive active returns and delivering incremental value over a passive benchmark.

These results reinforce the distinction between generating positive returns and delivering economically meaningful outperformance. While some strategies produce non-zero returns, the evidence for consistent and statistically robust excess performance is limited.

5.6. Feature Representation and Technical Indicators

Tables B3 and B4 examine whether feature construction improves performance. Table B3 shows that OHLC and candlestick representations produce broadly similar results. OHLC performs slightly better for long-only strategies, with an average Sharpe ratio of 0.026 compared to 0.020 for candlestick features. Candlestick features perform slightly better in long-short mode, but the difference is small.

By contrast, Table B4 shows a clearer benefit from extended features. Extended long-only strategies achieve an average Sharpe ratio of 0.047 and precision of 58.1%, compared with −0.002 and 52.3% for raw long-only features. Extended long-short strategies also outperform raw long-short strategies. This suggests that technical indicators add useful information, especially for selective long-side trading.

However, the benefit is not universal. Extended short-only strategies remain negative on average, despite higher precision. This again shows that better classification performance does not automatically translate into better trading performance.

5.7. Equity Curve Evidence

The equity curves in Appendix C support the numerical results. Strategies with strong headline Sharpe ratios generally show smoother cumulative performance, while high-turnover strategies are more vulnerable to sharp drawdowns and cost erosion. The contrast between no-cost and with-cost equity curves is especially important: transaction costs do not merely reduce final returns, but can change the entire shape of the performance path.

This visual evidence strengthens the interpretation from Tables A4 and B2. When trading signals are frequent and returns per trade are small, the strategy may look attractive in a frictionless backtest but become economically weak once costs are applied.

5.8. Overall Interpretation

Taken together, the results indicate that while ML models can extract predictive signals from cryptocurrency data, these signals are generally too small to generate robust economic profits once transaction costs are incorporated. The primary constraint is therefore not prediction accuracy, but the ability to convert signals into sufficiently large and infrequent trades to overcome trading frictions.

Long-only strategies exhibit greater robustness due to lower turnover, while extended features and GRU models provide improvements in selected cases. However, the overall evidence does not support a broad claim that machine learning-based active trading consistently outperforms passive benchmarks. Instead, performance is highly sensitive to asset characteristics, strategy design, and implementation frictions.

6. Conclusion, Limitations, and Future Work

6.1. Conclusion

This paper examines whether machine learning-based trading strategies outperform passive benchmarks in cryptocurrency markets once transaction costs are incorporated. The results show that this is only true to a limited extent.

While selected configurations achieve positive test Sharpe ratios, particularly for SOL and XRP (Table A2), active strategies do not consistently outperform buy-and-hold benchmarks (Table A1), and average performance remains modest after costs (Table A3). The key finding is that transaction costs materially reduce the economic value of machine learning trading strategies, especially those with high turnover.

Overall, the evidence indicates that although ML models can extract short-term predictive signals, these signals are generally too small to generate robust trading profits after transaction costs. The key constraint is therefore not prediction accuracy, but the ability to convert signals into economically meaningful trades after costs.

6.2. Limitations and Future Work

Several limitations qualify the interpretation of these results. First, the analysis relies on a fixed chronological train-validation-test split. While this provides a clean out-of-sample evaluation, it does not capture how performance would evolve under rolling or walk-forward retraining. Given the non-stationary nature of cryptocurrency markets, the reported results should therefore be interpreted as conditional on the chosen sample window rather than

evidence of persistent outperformance. Future work could adopt a walk-forward framework with periodic retraining to assess strategy stability across changing market regimes.

Second, the predictive task is formulated as a binary classification problem that focuses on return direction rather than magnitude. This creates a mismatch between model training and economic objectives, as correctly predicting small price movements is treated equivalently to large, economically meaningful changes. Although thresholding partially mitigates this issue, it does not fully align prediction with profitability. Future work could consider regression-based targets or directly optimize risk-adjusted return metrics. In addition, we do not explicitly account for potential class imbalance in directional labels, which may bias model predictions in trending markets.

Third, the backtesting framework, while incorporating transaction costs, remains stylized. Important frictions such as slippage, bid-ask spreads, market impact, and execution latency are not explicitly modeled. These factors would likely further reduce profitability, particularly for higher-turnover strategies. Incorporating richer execution assumptions would therefore provide a more conservative and realistic assessment of economic performance. Furthermore, the analysis uses a single transaction cost level, and does not examine sensitivity to alternative fee structures.

Finally, the use of hourly data introduces a trade-off between signal frequency and noise. While it enables an intraday trading design consistent with prior work, it also increases exposure to short-term noise and cost-driven performance erosion. Exploring lower-frequency strategies may improve signal quality and reduce turnover.

Appendix A: Core Empirical Results

All results are computed on the out-of-sample test period.

Table A1. Buy-and-Hold Benchmark Performance by Asset

Asset	Annualized Return	Annualized Volatility	Annualized Sharpe	Cumulative Return	Maximum Drawdown
BTC	0.309545	0.457861	0.818031	0.400965	-0.347615
ETH	0.131712	0.711194	0.530271	0.167299	-0.652824
XRP	1.416302	0.951227	**1.401183**	2.013191	-0.512126
SOL	-0.140333	0.857914	**0.252300**	-0.172253	-0.660623

Buy-and-hold performance on the test set.

Notes: Returns are computed using open-to-open price changes under a fully invested long position over the test period, with initial capital normalized to one. Annualized return and volatility are derived from hourly returns using a scaling factor of 8,760 hours per year, and the Sharpe ratio is computed assuming a zero risk-free rate. Maximum drawdown is defined as the largest peak-to-trough decline in cumulative returns.

Table A2. Best Feasible Active Strategy Configurations After Transaction Costs

Asset	Model	Model Family	Feature Set	Strategy Mode	Test Sharpe	Cumulative Return	Maximum Drawdown	Number of Trades	Annualized Sharpe
BTC	**GRU**	**ML**	**candle_extended**	**long_only**	**0.039497**	**0.014295**	**-0.169610**	**546.0**	**0.153603**

BTC	Logistic Regression	ML	ohlc_extended	long_short	-0.017681	-0.169962	-0.191816	788.0	-1.494822
BTC	Logistic Regression	ML	ohlc_extended	short_only	-0.025688	-0.179749	-0.196624	762.0	-1.730444
ETH	GRU	ML	ohlc_extended	long_only	-0.010586	-0.009783	-0.013672	50.0	-0.485698
ETH	**Logistic Regression**	**ML**	**candle_extended**	**long_short**	**0.043337**	**0.025087**	**-0.136006**	**329.0**	**0.219026**
ETH	Logistic Regression	ML	ohlc_raw	short_only	-0.012376	-0.325165	-0.394073	1528.0	-1.056474
SOL	GRU	ML	candle_extended	long_only	0.159148	0.176184	-0.107247	120.0	1.137860
SOL	**GRU**	**ML**	**candle_extended**	**long_short**	**0.199195**	**0.469984**	**-0.204527**	**228.0**	**1.791720**
SOL	GRU	ML	ohlc_extended	short_only	0.017333	-0.009314	-0.136910	314.0	0.049664
XRP	**GRU**	**ML**	**candle_extended**	**long_only**	**0.136439**	**0.273136**	**-0.123370**	**210.0**	**1.139927**
XRP	Logistic Regression	ML	ohlc_extended	long_short	-0.013537	-0.509166	-0.616527	1824.0	-1.041452
XRP	Logistic Regression	ML	ohlc_extended	short_only	-0.019984	-0.589651	-0.671322	1768.0	-1.503210

Best feasible configurations selected on validation data and evaluated on the test set.

Notes: Configurations are selected by first applying feasibility constraints on the validation set, requiring a minimum number of trades, a minimum signal ratio, and a minimum recall, and then choosing the threshold that maximizes validation precision among feasible candidates. Predicted probabilities are converted into trading signals using these fixed thresholds, and performance is evaluated on the test set. Returns are computed using open-to-open price changes with execution at the next-hour open and a one-hour holding period, consistent with the backtesting framework. Transaction costs of 0.035% per unit position change are applied, with reversals incurring 0.07%. Test Sharpe is computed from compounded signal returns as defined in Section 4, while annualized Sharpe is calculated from hourly net returns using a scaling factor of 8,760 hours per year. Number of trades reflects total executed trades over the test period.

Table A3. Average Feasible Performance by Model and Strategy Mode (With Cost)

Model	Strategy_Mode	Test Sharpe	Test Precision	Test Recall	Test Avg Return
GRU	**long_only**	**0.052970**	0.588544	0.084514	8.458190e-04
LightGBM	**long_only**	**0.015074**	0.548146	0.072491	3.792416e-04
Logistic Regression	**long_only**	**0.038226**	0.558170	0.038583	5.500315e-04
RSI	**long_only**	**-0.002637**	0.533457	0.451698	5.569275e-07
SMA	**long_only**	**0.026608**	0.523912	0.058512	3.623794e-04
XGBoost	**long_only**	**-0.012769**	0.517558	0.173099	-9.751999e-05
GRU	short_only	-0.055443	0.515812	0.091304	-3.747685e-04
LightGBM	short_only	-0.015110	0.507488	0.110456	-2.287981e-04

Logistic Regression	short_only	-0.004841	0.526865	0.331520	-4.327898e-05
RSI	short_only	-0.012719	0.516236	0.483814	-1.216415e-04
SMA	short_only	-0.023375	0.484834	0.306747	-5.030647e-04
XGBoost	short_only	-0.024454	0.503124	0.199777	-2.064850e-04
GRU	long_short	0.029658	0.577249	0.094824	2.174115e-04
LightGBM	long_short	-0.009592	0.531893	0.089839	-1.100359e-04
Logistic Regression	long_short	0.007647	0.556135	0.138734	8.927192e-05
RSI	long_short	-0.009022	0.524681	0.463395	-9.669564e-05
SMA	long_short	-0.012337	0.493434	0.123191	-1.323381e-04
XGBoost	long_short	-0.019610	0.510700	0.192056	-2.012464e-04

Average performance across feasible configurations by model and strategy mode.

Notes: Values are averaged across all assets, feature sets, and feasible configurations that satisfy the validation-stage constraints on minimum trades, signal ratio, and recall. Each configuration corresponds to a fixed model, feature set, and threshold selected on the validation set. Test Sharpe is computed from compounded signal returns as defined in Section 4, while test precision and recall are evaluated on executed trades. Test Avg Return denotes the mean realized return per active trade after transaction costs. All metrics are reported under the with-cost regime.

Table A4. Average Transaction Cost Drag by Model and Strategy Mode

Model	**Strategy Mode**	**Sharpe Cost Drag**	**Avg Return Cost Drag**	**Cumulative Return Cost Drag**

GRU	long_only	0.024460	0.000209	0.179548
GRU	long_short	0.019931	0.000161	0.257989
GRU	short_only	0.028632	0.000179	0.236016
LightGBM	long_only	0.038513	0.000274	0.321102
LightGBM	long_short	0.029821	0.000242	0.514533
LightGBM	short_only	0.031328	0.000249	0.320139
Logistic Regression	long_only	0.030738	0.000275	0.261144
Logistic Regression	long_short	0.025920	0.000177	0.524344
Logistic Regression	short_only	0.024100	0.000150	0.495159
RSI	long_only	0.009495	0.000068	0.589230
RSI	long_short	0.007527	0.000088	0.578891
RSI	short_only	0.005626	0.000059	0.331985
SMA	long_only	0.002002	0.000017	0.030211
SMA	long_short	0.001209	0.000014	0.038244
SMA	short_only	0.000433	0.000009	0.021534
XGBoost	long_only	0.030239	0.000237	0.641368
XGBoost	long_short	0.025520	0.000256	0.975676
XGBoost	short_only	0.025738	0.000230	0.588234

Average reduction in performance due to transaction costs.

Notes: Cost drag is defined as the difference between performance computed under the no-cost and with-cost regimes, using identical model configurations and thresholds. All values are averaged across assets, feature sets, and feasible configurations. Sharpe cost drag is computed as

the difference in test Sharpe, while average return and cumulative return cost drag are computed from net realized returns. All metrics are based on open-to-open returns under the execution assumptions described in Section 4.

Table A5. Bootstrap-Based Statistical Usefulness of Best Strategies

Asset	Model	Strategy Mode	Mean Net Return	p-value (vs 0)	Mean Excess Return (vs Buy-and-Hold)	p-value (vs Buy-and-Hold)
BTC	GRU	long_only	2.202891e-06	0.4403	-0.000043	0.8425
BTC	Logistic Regression	long_short	-1.651581e-05	0.9581	-0.000060	0.8948
BTC	Logistic Regression	short_only	-1.767732e-05	0.9776	-0.000061	0.8990
ETH	GRU	long_only	-8.864273e-07	0.7394	-0.000048	0.7441
ETH	Logistic Regression	long_short	3.208571e-06	0.4136	-0.000040	0.6974
ETH	Logistic Regression	short_only	-3.194912e-05	0.8689	-0.000075	0.7945
SOL	GRU	long_only	1.557209e-05	0.0878	-0.000013	0.5642
SOL	GRU	long_short	3.705404e-05	0.0106	0.000009	0.4647
SOL	GRU	short_only	1.020538e-06	0.4813	-0.000027	0.6211
XRP	GRU	long_only	2.406502e-05	0.0709	-0.000134	0.9233

XRP	Logistic Regression	long_short	-5.349166e-05	0.9091	-0.000207	0.9593
XRP	Logistic Regression	short_only	-7.155204e-05	0.9614	-0.000225	0.9694

Bootstrap-based tests mean strategy returns and excess returns.

Notes: Mean net return is computed as the average realized return per period under the with-cost regime. Mean excess return is defined as the difference between the strategy return and the corresponding buy-and-hold return for the same asset. Statistical significance is evaluated using bootstrap resampling of the return series, with p-values reported for the null hypotheses of zero mean return and zero excess return, respectively. All tests are conducted on the out-of-sample test period using the same return definitions as in Section 4.

Appendix B. Supporting Results

All results are computed on the out-of-sample test period under the with-cost regime unless otherwise stated.

Table B1. Sample of Feasible Strategy Configurations (With Cost)

Asset	Model	Feature Set	Strategy Label	Annualized Sharpe	Cumulative Return	Maximum Drawdown	Number of Trades
XRP	Logistic Regression	candle_extended	long	2.264394	0.705594	-0.158772	318.0
SOL	GRU	candle_extended	long_short	1.791720	0.469984	-0.204527	228.0
SOL	Logistic Regression	candle_extended	long	1.663648	0.592129	-0.154548	490.0
XRP	LightGBM	candle_extended	long	1.662529	0.568387	-0.145782	174.0
XRP	Logistic Regression	ohlc_extended	long	1.554226	0.670138	-0.189672	428.0
ETH	LightGBM	candle_raw	short	1.538350	0.279370	-0.064289	504.0
XRP	SMA	rule_based	long	1.526516	1.124256	-0.336444	92.0
BTC	GRU	ohlc_raw	long_short	1.439082	0.242300	-0.070163	114.0
XRP	GRU	candle_extended	long	1.139927	0.273136	-0.123370	210.0
SOL	GRU	candle_extended	long	1.137860	0.176184	-0.107247	120.0

Representative subset of feasible configurations.

Notes: This table reports a subset of feasible configurations selected for illustration. Feasible configurations satisfy the validation-stage constraints on minimum trades, signal ratio, and recall. Metrics are computed on the test set using the same definitions as in Appendix A.

Table B2. Comparison of Strategy Performance Before and After Transaction Costs

Asset	Model	Strategy Mode	Test Sharpe no_cost	Test Sharpe with_cost	Cumulative Return no_cost	Cumulative Return with_cost	Sharpe Cost Drag	Return Cost Drag
BTC	LightGBM	long_only	0.194980	0.094050	0.077686	-0.001470	0.100930	0.079155
BTC	LightGBM	long_only	0.026888	-0.068100	0.034784	-0.196877	0.094988	0.231661
BTC	LightGBM	long_short	0.161201	0.072847	0.189135	-0.009401	0.088354	0.198536
ETH	GRU	long_only	0.075632	-0.010586	0.007699	-0.009783	0.086218	0.017483
BTC	LightGBM	short_only	0.095519	0.015849	0.252769	-0.133688	0.079671	0.386457
BTC	LightGBM	long_short	0.065511	-0.008619	0.221080	-0.226909	0.074130	0.447988
BTC	LightGBM	short_only	0.077563	0.007114	0.158444	-0.111510	0.070449	0.269954
BTC	GRU	short_only	0.076940	0.008111	1.074633	-0.402811	0.068829	1.477444
BTC	Logistic Regression	short_only	0.084394	0.017440	0.436525	-0.172543	0.066954	0.609068
ETH	GRU	short_only	-0.313806	-0.379498	-0.031640	-0.045115	0.065693	0.013476

Selected top 10 configurations ranked by Sharpe cost drag.

Notes: Each row corresponds to a fixed model, feature set, and threshold configuration evaluated under both no-cost and with-cost settings. The table reports the ten configurations with the largest reduction in test Sharpe when transaction costs are introduced. Cost drag is defined as the difference between no-cost and with-cost performance for the same configuration. Test Sharpe is computed from compounded signal returns, and cumulative return reflects total compounded performance over the test period. All metrics use the same return definitions and execution assumptions as described in Section 4.

Table B3. Input Representation Comparison (OHLC vs Candlestick)

Input Type	Strategy Mode	Annualized Sharpe	Test Precision	Average Return
ohlc	long_only	0.025516	0.562904	0.000425
candle	long_only	0.019519	0.542504	0.000401
candle	long_short	0.004694	0.566658	0.000043
ohlc	long_short	-0.000642	0.521330	-0.000046
candle	short_only	-0.013162	0.515034	-0.000171
ohlc	short_only	-0.036762	0.511610	-0.000256

Performance comparison across input representations.

Notes: Results are averaged across all assets and feasible configurations. Metrics are computed under the with-cost regime using the same definitions as in Appendix A.

Table B4. Feature Engineering Comparison (Raw vs Extended Features)

Feature Type	Strategy Mode	Annualized Sharpe	Test Precision	Average Return
extended	long_only	0.047231	0.580818	0.000809
raw	long_only	-0.002390	0.523354	0.000003
extended	long_short	0.011195	0.576018	0.000113
raw	long_short	-0.007143	0.511970	-0.000115
raw	short_only	-0.020162	0.495822	-0.000180
extended	short_only	-0.029762	0.530823	-0.000247

Performance comparison across feature sets.

Notes: Results are averaged across all assets and feasible configurations. Extended features include technical indicators, while raw features consist of price and volume inputs. All metrics are computed under the with-cost regime.

Appendix C. Equity Curve Analysis

Figure C1-C4: Best Strategy vs Buy-and-Hold (With Cost)

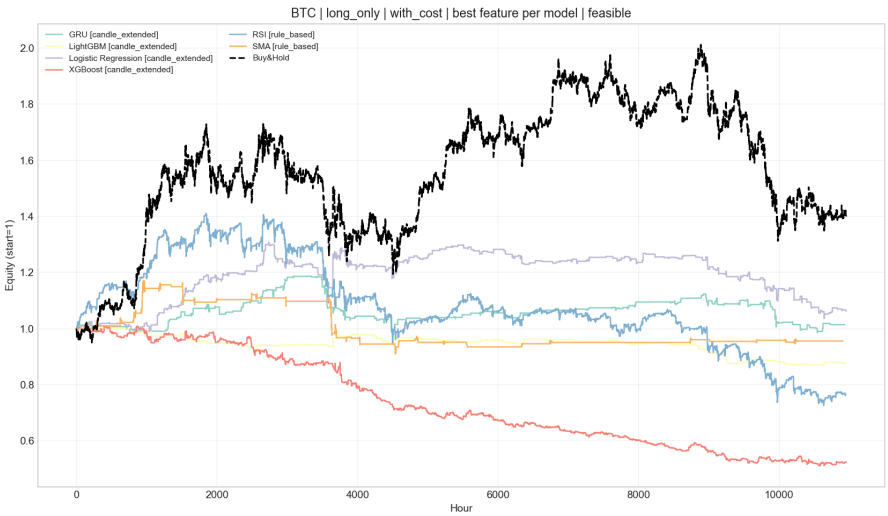


Figure C1. BTC Long Only (With Cost) Best Feature Per Model

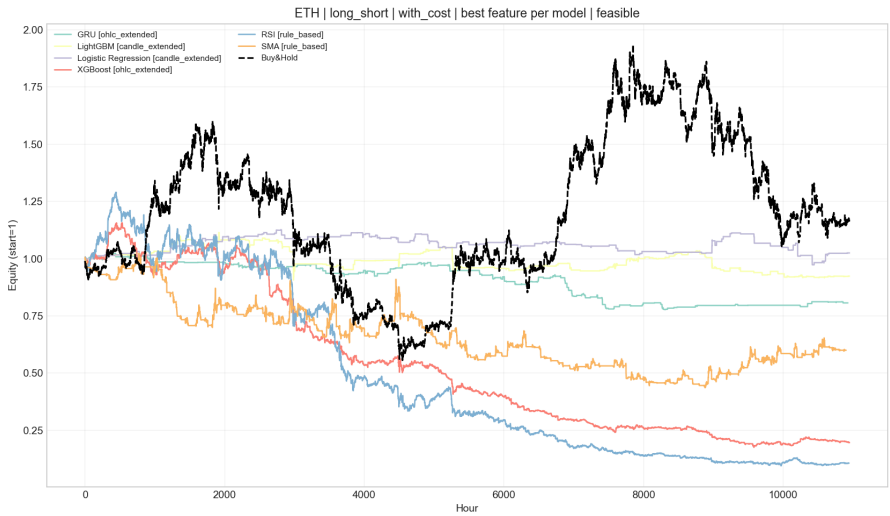


Figure C2. ETH Long Short (With Cost) Best Feature Per Model

Figure C3. SOL Long Short (With Cost) Best Feature Per Model

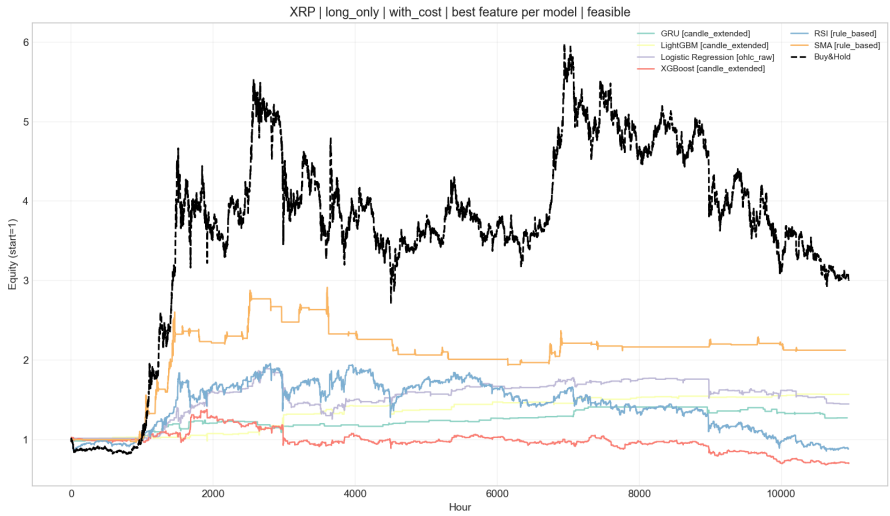
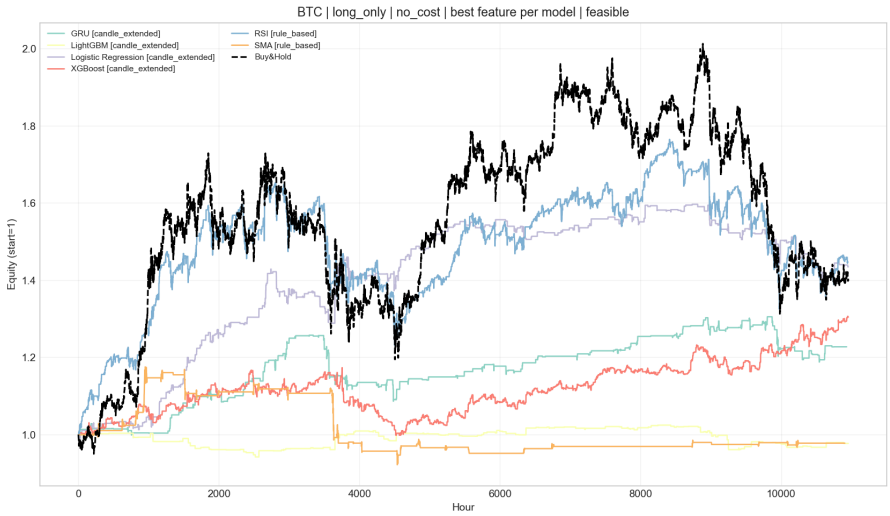


Figure C4. XRP Long Only (With Cost) Best Feature Per Model

The selected strategies correspond to the highest-performing feasible configurations reported in Table A2. While certain strategies exhibit periods of outperformance, buy-and-hold remains competitive and more stable over the full test period.

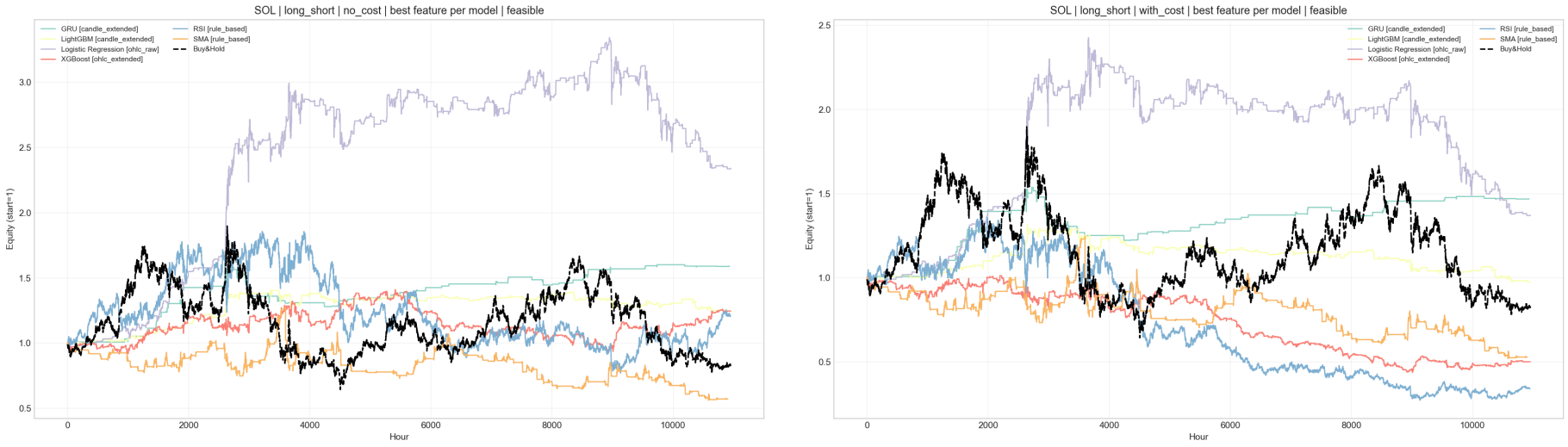
Figure C5-C6: Effect of Transaction Costs



No Cost

With Cost

Figure C5. BTC Long Only Cost Comparison



No Cost

With Cost

Figure C6. SOL Long Short Cost Comparison

Transaction costs substantially reduce cumulative returns, particularly for strategies with higher turnover. This highlights the gap between frictionless backtest performance and realistic trading outcomes.

Appendix D. Regime-Shift Robustness Analysis

This appendix presents a robustness analysis based on a shifted sample window (2020-2024), compared to the main analysis period (2021-2025). The objective is to assess whether the performance of machine learning and rule-based trading strategies is stable across different market environments.

The analysis re-runs the full modelling, validation, and backtesting pipeline under identical specifications, including feature construction, model training, threshold tuning, and transaction cost assumptions. This ensures that differences in results are attributable solely to changes in the underlying market conditions rather than methodological variation.

The results are reported as direct comparisons between the two sample windows.

Table D1. Strategy Performance by Mode Across Sample Windows

Window	Strategy Mode	Cost Regime	Avg Test Sharpe	Avg Cumulative Return	Avg Number of Trades	Positive Sharpe Share
2020-2024	long_only	with_cost	0.048408	0.029204	1119	0.736111
2021-2025	long_only	with_cost	0.021228	-0.106543	1068	0.500000
2020-2024	long_short	with_cost	0.017134	-0.209982	1908	0.472222
2021-2025	long_short	with_cost	0.000614	-0.421724	2144	0.263889
2020-2024	short_only	with_cost	-0.024126	-0.432444	1382	0.138889
2021-2025	short_only	with_cost	-0.024194	-0.405698	1564	0.194444

2020-2024	long_only	no_cost	0.076687	0.460417	1119	0.902778
2021-2025	long_only	no_cost	0.049492	0.239684	1068	0.791667
2020-2024	long_short	no_cost	0.043308	0.385085	1908	0.777778
2021-2025	long_short	no_cost	0.023586	0.117570	2144	0.625000
2020-2024	short_only	no_cost	-0.003592	-0.140453	1382	0.402778
2021-2025	short_only	no_cost	0.000543	-0.021713	1564	0.458333

Comparison of average test Sharpe ratios (with cost) across strategy modes for 2021-2025 and 2020-2024 sample windows.

Values are averaged across all feasible configurations by strategy mode. Metrics are computed under the with-cost regime using the same definitions as in Appendix A. The comparison isolates the effect of sample period differences on strategy performance.

Table D2. Strategy Performance by Model Family Across Sample Windows

Window	**Model Family**	**Cost Regime**	**Avg Test Sharpe**	**Avg Cumulative Return**	**Avg Number of Trades**
2020-2024	Rule	with_cost	-0.004956	-0.167128	1062
2021-2025	Rule	with_cost	-0.005580	-0.353686	1077
2020-2024	Rule	no_cost	0.001091	0.193171	1062
2021-2025	Rule	no_cost	-0.001198	-0.088670	1077
2020-2024	ML	with_cost	0.016151	-0.209067	1520
2021-2025	ML	with_cost	-0.000410	-0.306026	1656
2020-2024	ML	no_cost	0.043515	0.240247	1520

2021-2025	ML	no_cost	0.027529	0.136912	1656

Comparison of average test Sharpe ratios (with cost) across model families for 2021-2025 and 2020-2024 sample windows.

Results are averaged across all assets, feature sets, and feasible configurations. Model families include machine learning models (Logistic Regression, XGBoost, LightGBM, GRU) and rule-based strategies (SMA, RSI).

Table D3. Buy-and-Hold Performance Comparison Across Sample Windows

Window	Asset	Cumulative Return	Annualized Return	Annualized Sharpe	Maximum Drawdown
2021-2025	BTC	0.400965	0.309545	0.818031	-0.347615
2020-2024	BTC	2.299935	1.598082	2.147217	-0.323107
2021-2025	ETH	0.167299	0.131712	0.530271	-0.652824
2020-2024	ETH	0.930191	0.691986	1.167748	-0.466551
2021-2025	SOL	-0.172253	-0.140333	0.252300	-0.660623
2020-2024	SOL	2.342186	2.001117	1.635658	-0.465184
2021-2025	XRP	2.013191	1.416302	1.401183	-0.512126
2020-2024	XRP	2.967493	2.010497	1.715271	-0.457828

Comparison of buy-and-hold cumulative returns across assets for 2021-2025 and 2020-2024 sample windows.

Returns are computed under a fully invested long position. Differences reflect variation in underlying market conditions rather than strategy design.

Table D4. Statistical Significance Summary Across Sample Windows

Window	Grouped Tests Significant (p<0.05)	Grouped Tests Total	Usefulness Significant vs 0	Usefulness Total	Beat Buy-and-Hold Significant
2021-2025	11	90	4	24	0
2020-2024	23	90	9	24	0

Summary of bootstrap-based statistical tests across sample windows.

The table reports the number of configurations with statistically significant mean returns (vs zero) and excess returns (vs buy-and-hold). Significance is evaluated using bootstrap resampling at conventional levels.

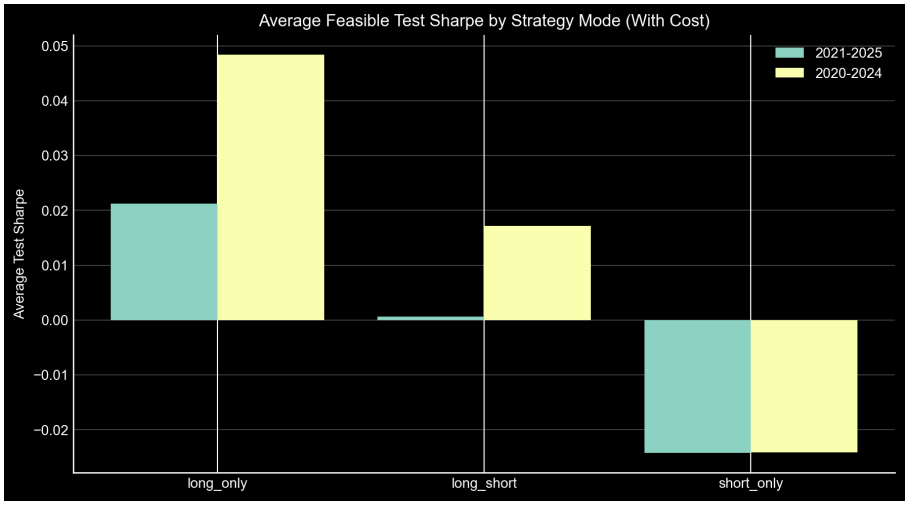


Figure D1. Strategy Mode Performance (With Cost)

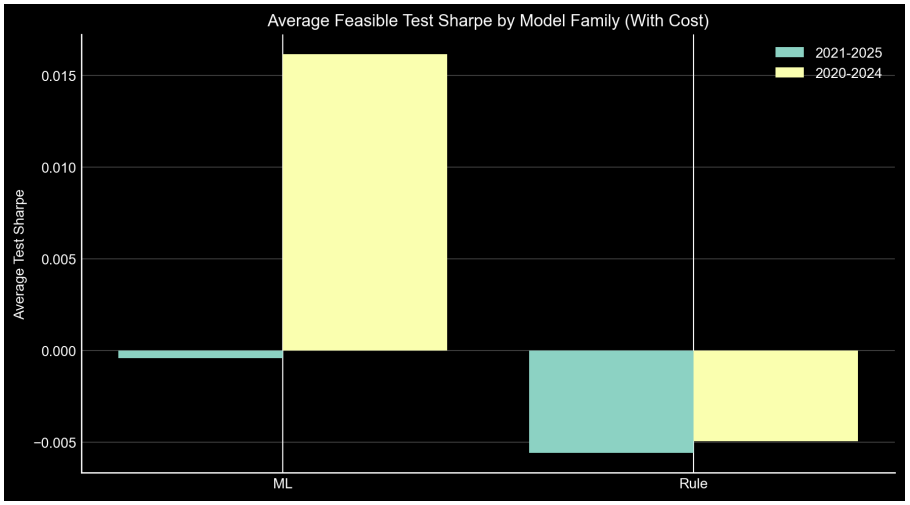


Figure D2. Model Family Performance (With Cost)

Figure D3. Buy-and-Hold Cumulative Return Comparison

Overall, the regime-shift analysis shows that while absolute performance improves in the 2020-2024 sample, particularly for long-only and long-short strategies, the relative conclusions remain unchanged. Transaction costs continue to materially reduce performance, short-only strategies remain weak, and machine learning models do not deliver statistically significant outperformance over buy-and-hold benchmarks.

This suggests that the main findings of the report are robust to changes in sample window, and that improvements in favourable market conditions reflect broader market trends rather than persistent strategy-level alpha.

Rebuttal to Reviewer Comments

We thank the reviewers for their constructive feedback. In response, we have revised the report to sharpen the research question, substantially strengthen the Results and Analysis section, and clarify methodological choices and limitations.

Specifically, we strengthen the discussion of key results using direct references to Tables A1-A5 and supporting appendix tables, and expand the interpretation of strategy behaviour, including the underperformance of short-only strategies, the role of turnover, and the comparison with buy-and-hold benchmarks. In addition, we incorporate equity-curve analysis and a regime-shift robustness appendix to provide a more comprehensive and data-driven evaluation.

We agree that a single chronological train-validation-test split cannot establish cross-regime generalisation. This limitation is now explicitly stated in Section 6. While the sample spans multiple market phases, the results should be interpreted as conditional on the selected window rather than evidence of persistent outperformance. To further address this concern, we include an additional robustness check by re-running the full pipeline on an alternative sample period (2020-2024). The results, reported in Appendix D, show that while performance levels vary across periods, the core conclusion remains unchanged: ML strategies do not deliver consistent, cost-robust outperformance across different market environments.

On regime-conditioned analysis, we acknowledge that a formal regime classification would provide a more granular breakdown of performance. Instead, we adopt the shifted-window robustness approach described above, which preserves the full modelling and backtesting framework and allows a consistent comparison across periods.

For the binary directional target, we agree that predicting direction ignores return magnitude and creates a mismatch with economic objectives. In our framework, this is

partially mitigated by validation-stage thresholding and evaluation using Sharpe ratio, cumulative return, drawdown, and transaction-cost drag. However, we acknowledge that this does not fully align training with profitability, and we now explicitly include regression-based targets and direct optimisation of risk-adjusted returns as future work in Section 6.

We clarify the Sharpe ratio definitions in Section 4.4 to address concerns about multiple formulations. The report now distinguishes between the paper-style signal-return Sharpe used as the primary evaluation metric and the annualised hourly Sharpe used as a secondary diagnostic, ensuring consistent interpretation across all reported results.

On backtesting realism, we agree that the framework remains stylised. While transaction costs are incorporated, other frictions such as slippage, execution delay, market impact, and short-selling constraints are not explicitly modelled. These limitations are now clearly stated in Section 6, and we emphasise that the reported results should be interpreted as more realistic than frictionless backtests but still optimistic relative to a fully implementable trading environment.

We acknowledge the suggestion to conduct transaction cost sensitivity analysis across multiple fee levels. In this study, we adopt a single cost assumption of 0.035%, reflecting typical trading fees on major exchanges such as Binance. Rather than performing a full cost sweep, we compare no-cost and with-cost settings to isolate the impact of trading frictions. The revised Results section now explicitly demonstrates that even at this relatively low cost level, transaction costs materially reduce performance and often eliminate profitability, indicating that they are already a binding constraint.

Regarding the precision-first thresholding rule, we clarify that model training and trading evaluation are deliberately separated. Models are trained to identify directional signals, while validation thresholds are used to select higher-confidence trades subject to

feasibility constraints. Economic performance is then evaluated using Sharpe ratio and return-based metrics. This design avoids overfitting the trading rule to a single validation period, although direct optimisation of economic metrics remains a valid extension.

For the use of hourly data, we agree that high-frequency returns are noisy and can generate excessive turnover. We retain the hourly horizon to remain aligned with the reference study and to evaluate intraday strategies, but now explicitly acknowledge in Section 6 that lower-frequency data may improve signal quality and reduce cost-driven performance erosion.

We address concerns regarding feature redundancy by noting that while inputs may be correlated, the dimensionality is moderate and features are intentionally retained to allow comparison across raw, candlestick, and extended technical-indicator representations. We therefore do not apply aggressive feature reduction in the main analysis.

We clarify that the comparison between ML and rule-based strategies is not perfectly symmetric due to differences in model flexibility. ML models learn predictive relationships from data, while SMA and RSI rely on predefined indicator structures with validation-tuned parameters and thresholds. However, both classes apply validation-based threshold selection before test evaluation, ensuring that the trading rule is applied consistently across strategy families.

Finally, we acknowledge that the report does not include a full feature-importance or explainability analysis. As the primary objective of this project is economic evaluation rather than model interpretability, we focus on comparing feature representations empirically in the Results section. Explainability methods such as SHAP or permutation importance are identified as future work.

References

Goutte, S., Le, H., Liu, F., & Mettenheim, H. (2023). Deep learning and technical analysis in cryptocurrency market. *Finance Research Letters. Vol 54, June 2023, 103809.* https://doi.org/10.1016/j.frl.2023.103809

Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies, 33(5), 2223–2273*. https://doi.org/10.1093/rfs/hhaa009